

ELECTRONIC VOTING MACHINE IN VISUAL BASIC

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The electronic voting machine (EVM) implemented in several states by the Election Commission of India is a novel method over the traditional ballot-box voting system. It ensures privacy and security needed to conduct the polls. If recent trends in software and hardware are put to use, we may get more sophisticated EVMs that make polling as simple as e-banking or ATM-based transactions.

Here we present an EVM that is purely software based. The facility for 'mock poll,' which is usually conducted prior to the actual polling in order to check the accuracy of the EVM, is included in the program. You can clear the mock-poll result after testing the EVM's functioning.

The EVM comprises two separate units: control unit and ballot unit.

The control unit has a keyboard to enter the voter's serial number and special ID number, and a screen for displaying the voter's photo (refer Screenshot 1). 'Power on' and 'exit' buttons are provided to start and exit the program. The control unit also has the facility to display the instant result and candidate-wise result at the bottom of the screen with the help of start, stop, result and total buttons.

The ballot unit has display windows for candidates' symbols, serial numbers and names. After casting your vote, the display on EVM appears as shown in Screenshot 2.

A special ID number, akin to signature, is allotted to every voter. The polling officials identify a voter by observing on the screen his displayed photo and voter ID card number just by entering his serial number in the electoral roll. Thereafter, the voter is made to enter his special ID number.

If the voter qualifies through these two steps, he is allowed to cast his vote.

The program has the facility to check total voting, including male and

female separately, at any time by entering a password known to the returning officer. The program also has the facility to display current time along with the voting result as it is desirable to know the result on hourly basis.

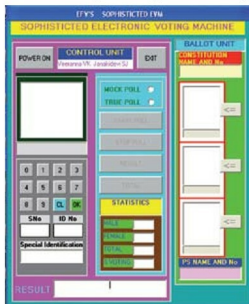
To keep the program short and simple, it is limited to the display of three candidates (Congress, BJP and Samajwadi) and results of a single polling station. You cannot add more candidates here but the program can be extended to accommodate more candidates by changing the form design and program code. Only four voters' details are included in this program. In order to add the details of more voters, you have to open MS-Access application and update the database. (Normally, data entry is done at the returning officer's end.)

The program is tested under Windows XP, MS Access and VB6 environment, with all files kept in 'E:\EFYEVM' folder. It uses ADODB (ActiveX data control object), RDO (remote data object) and MS Access database.

Designing of electronic voting machine

Form design. The program is based on a single form. Open Visual Basic (VB) and select 'EXE project' to open VB form1. Change the properties of form1 as follows:

Caption	- EFY's sophisticated EVM
FillStyle	- 0-solid
StartPosition	- 2-center of screen



Screenshot 1: Program output of EVM



Screenshot 2: Mock poll result